

2023 AMC 10B Problem 15 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 15. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 15

**Problem:**

What is the least positive integer m such that $m \cdot 2! \cdot 3! \cdot 4! \cdot 5! \cdots 16!$ is a perfect square?

Answer Choices:

- A. 30
- B. 30,030
- C. 70
- D. 1430
- E. 1001

Solution:

Because $n! \cdot (n+1)! = n! \cdot n! \cdot (n+1) = (n!)^2 \cdot (n+1)$, the given product can be rewritten as

$$m \cdot 2 \cdot (3!)^2 \cdot 4 \cdot (5!)^2 \cdot 6 \cdots (15!)^2 \cdot 16$$

Therefore it suffices to make

$$m \cdot 2 \cdot 4 \cdot 6 \cdot 8 \cdot 10 \cdot 12 \cdot 14 \cdot 16 = m \cdot 2^{15} \cdot 3^2 \cdot 5 \cdot 7$$

a perfect square. The least possible value of m is $2 \cdot 5 \cdot 7 = (C) \boxed{70}$.

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